pressing questions and thorniest puzzles on which the standard big bang model is mute.

Heat and Symmetry

When things get very hot or very cold, they sometimes change. And sometimes the change is so pronounced that you can't even recognize the things with which you began. Because of the torrid conditions just after the bang, and the subsequent rapid drop in temperature as space expanded and cooled, understanding the effects of temperature change is crucial in grappling with the early history of the universe. But let's start simpler. Let's start with ice.

If you heat a very cold piece of ice, at first not much happens. Although its temperature rises, its appearance remains pretty much unchanged. But if you raise its temperature all the way to 0 degrees Celsius and you keep the heat on, suddenly something dramatic does happen. The solid ice starts to melt and turns into liquid water. Don't let the familiarity of this transformation dull the spectacle. Without previous experiences involving ice and water, it would be a challenge to realize the intimate connection between them. One is a rock-hard solid while the other is a viscous liquid. Simple observation reveals no direct evidence that their molecular makeup, H_2O, is identical. If you'd never before seen ice or water and were presented with a vat of each, at first you would likely think they were unrelated. And yet, as either crosses through 0 degrees Celsius, you'd witness a wondrous alchemy as each transmutes into the other.

If you continue to heat liquid water, you again find that for a while not much happens beyond a steady rise in temperature. But then, when you reach 100 degrees Celsius, there is another sharp change: the liquid water starts to boil and transmute into steam, a hot gas that again is not obviously connected to liquid water or to solid ice. Yet, of course, all three share the same molecular composition. The changes from solid to liquid and liquid to gas are known as *phase transitions.* Most substances go through a similar sequence of changes if their temperatures are varied through a wide enough range.[1]